CMPG 325 – COMPUTER NETWORKS

FINAL TECHNICAL REPORT

Network Design, Subnetting, Router-on-a-Stick Implementation, DNS Challenge, and Public Kiosk Security (CR13)

Client Organization	JB Marks Local Municipality Satellite Office (Ventersdorp)
Client ID	CLI-105
Project ID	CMPG325-2026-105
Industry Sector	Municipal Services & Local Government
Assigned Addressing Block	172.30.72.0/23
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Course Module	CMPG 325 – Computer Networks

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1. Introduction

This report documents the complete design, implementation, and testing of a computer network for the JB Marks Local Municipality Satellite Office in Ventersdorp. The network was designed using the assigned addressing block 172.30.72.0/23 and fulfills all client requirements, including:

- **Internal DNS Name Resolution:** Deployment of an internal Domain Name System (DNS) server for municipal name resolution across all subnets.
- **VoIP Traffic Isolation:** Strict hardware and logical separation of Voice over IP (VoIP) traffic on VLAN 30 to guarantee Quality of Service (QoS).
- **Public Kiosk Security (CR13):** Deployment of a reception self-service terminal governed by an extended Access Control List (ACL) granting public internet access while strictly prohibiting reachability to internal municipal networks.

The network was implemented in Cisco Packet Tracer using a Hierarchical Star topology with Router-on-a-Stick inter-VLAN routing. All configurations have been rigorously tested and verified across multiple test cycles.

2. Client Requirements Analysis

2.1 Client Overview

The client is the JB Marks Local Municipality Satellite Office located in Ventersdorp. Operating within the Municipal Services sector, this office requires a resilient, secure, and structured network infrastructure to facilitate day-to-day administrative tasks, inter-departmental collaboration, server hosting, IP telephony, and public-facing self-service facilities.

2.2 Key Requirements

- **End-to-End Municipal Connectivity:** The network infrastructure must establish full connectivity and routing across all operational municipal departments.
- **VoIP Traffic Separation:** Voice traffic generated by Cisco IP phones must be isolated on a dedicated voice VLAN to prevent packet jitter, latency, and call degradation.
- **Internal DNS Name Resolution:** An internal DNS service must resolve local hostnames (e.g., printer, servers, gateways) into their respective IP addresses for staff convenience.
- **Public Kiosk Isolation (CR13):** As mandated by Change Request CR13, a public self-service kiosk situated at reception must receive automated IP leasing and internet connectivity, but MUST be completely blocked from reaching internal municipal subnets.

2.3 Addressing Constraint

The organization was assigned the IP addressing block 172.30.72.0/23 (providing 512 total addresses across 172.30.72.0 – 172.30.73.255). All departmental subnets, gateways, and end-device pools must be derived exclusively from this block through structured Variable Length Subnet Masking (VLSM).

3. Network Design

3.1 Physical Topology

The network utilizes a Hierarchical Star with Centralized Routing topology (Router-on-a-Stick). This design connects a central gateway Router to a Core Switch via an 802.1Q trunk link, which in turn aggregates connections to five departmental Access Switches.

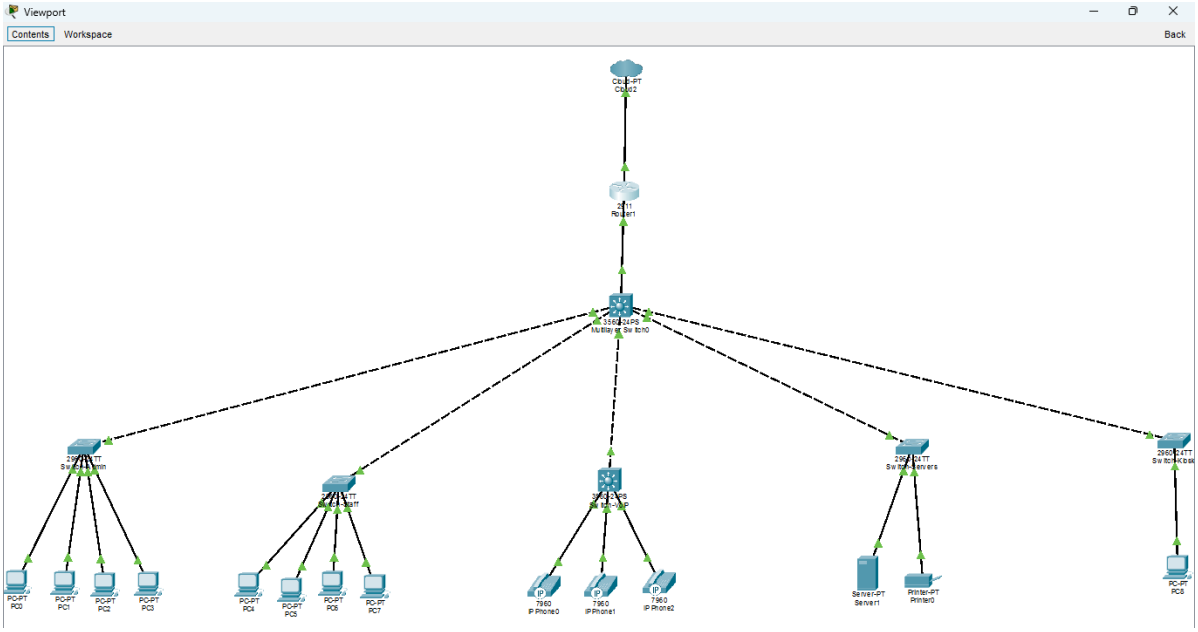


Figure 1: Full physical network topology with active 802.1Q trunks and departmental access switches in Cisco Packet Tracer

3.2 Logical Topology

To optimize network traffic and enforce security perimeters, the 172.30.72.0/23 block was segmented into 5 logical VLANs using custom subnet masks based on departmental device density and security requirements.

3.3 IP Addressing Plan

The detailed IP addressing allocation plan is presented below:

Department	VLAN	Network Address	CIDR	Subnet Mask	Usable Range	Gateway
Administration	VLAN 10	172.30.72.0	/25	255.255.255.128	172.30.72.1 – .126	172.30.72.1
Staff	VLAN 20	172.30.72.128	/25	255.255.255.128	172.30.72.129 – .254	172.30.72.129
VoIP (Phones)	VLAN 30	172.30.73.0	/26	255.255.255.192	172.30.73.1 – .62	172.30.73.1
Servers	VLAN 40	172.30.73.64	/28	255.255.255.240	172.30.73.65 – .78	172.30.73.65
Kiosk (CR13)	VLAN 50	172.30.73.80	/28	255.255.255.240	172.30.73.81 – .94	172.30.73.81

Static Server Infrastructure: Key Static Server Assignments:

- **Central DHCP & DNS Server:** 172.30.73.66 / 255.255.255.240 (Gateway: 172.30.73.65)
- **Shared Network Printer:** 172.30.73.68 / 255.255.255.240 (Gateway: 172.30.73.65)

3.4 Device Inventory

The equipment inventory deployed across the satellite office is detailed below:

Device Type	Model / Spec	Quantity	Operational Purpose
Main Router	Cisco 2811	1	Central gateway, inter-VLAN routing, CME telephony & ACL policy enforcement
Core Switch	Cisco 3560-24PS	1	Central distribution switch aggregating 802.1Q trunks from departmental switches
Access Switches	Cisco 2960-24TT	4	Standard FastEthernet access connectivity for Admin, Staff, Servers, and Kiosk
VoIP Access Switch	Cisco 3560-24PS	1	Power-over-Ethernet (PoE) access switch providing power and voice VLAN support
Services Server	Server-PT	1	Hosts centralized multi-pool DHCP leasing and municipal internal DNS services
Network Printer	Printer-PT	1	Centralized network printing resource on VLAN 40
Admin Workstations	PC-PT	4	Administrative desktop PCs assigned to VLAN 10
Staff Workstations	PC-PT	4	General staff desktop PCs assigned to VLAN 20
IP Telephones	Cisco 7960	3	VoIP handsets registered with CME on VLAN 30
Public Kiosk PC	PC-PT	1	Reception self-service PC locked to VLAN 50 with internet-only access

Implementation Note: Note on Hardware Selection: The router hardware was upgraded from a Cisco 2911 to a Cisco 2811 during implementation because the 2811 IOS image in Packet Tracer provides full Call Manager Express (CME) feature set support required for ephone registration and extension assignment.

3.5 Design Justification

3.5.1 Hierarchical Star Architecture

The Hierarchical Star topology concentrates routing decisions at the Cisco 2811 Router while offloading Layer 2 switching to the Cisco 3560 Core Switch. This separates collision domains, simplifies troubleshooting, and permits modular expansion—allowing new departmental access switches to connect directly to the Core Switch without disrupting existing subnets.

3.5.2 Subnet Separation and Security Boundaries

- **VoIP Quality of Service:** Isolating voice traffic on VLAN 30 protects time-sensitive voice packets from data bursts, preventing jitter, latency, and dropped calls.
- **Public Kiosk Security (CR13):** Isolating the reception terminal on VLAN 50 allows applying extended ACL 100 on the router sub-interface, permitting WAN traffic while blocking access to internal subnets.
- **Server Infrastructure Protection:** Hosting DNS/DHCP infrastructure on VLAN 40 restricts administrative access and shields core municipal services from workstation broadcast noise.

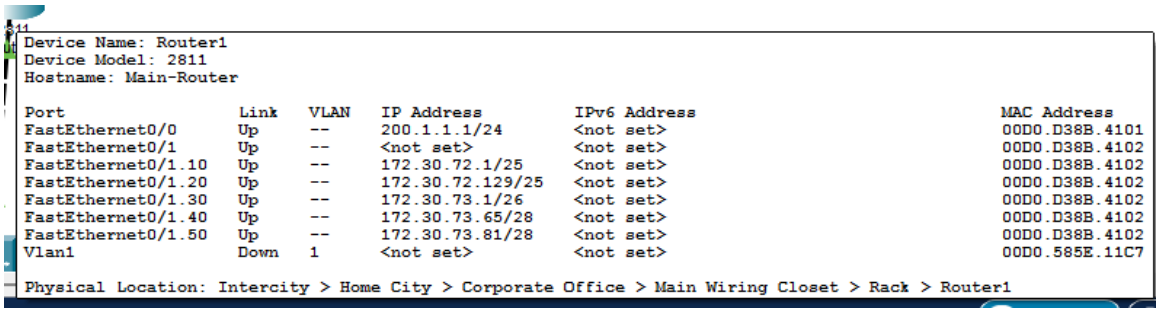
3.5.3 VLSM Subnet Sizing Justification

- **Admin & Staff (/25):** /25 subnets (126 usable hosts) provide ample growth room for staff expansion.
- **VoIP VLAN (/26):** /26 subnet (62 usable hosts) accommodates telephony hardware needs for satellite operations.
- **Servers & Kiosk (/28):** /28 subnets (14 usable hosts each) prevent IP address waste for restricted host groups.

4. Implementation & Configuration

4.1 Router Configuration

The Cisco 2811 Router was configured with 802.1Q sub-interfaces on FastEthernet0/1 to execute inter-VLAN routing. The ip helper-address 172.30.73.66 directive was configured on sub-interfaces 10, 20, 30, and 50 to relay DHCP broadcast requests across subnets to the central server.



```
Device Name: Router1
Device Model: 2811
Hostname: Main-Router

Port                Link   VLAN   IP Address          IPv6 Address                              MAC Address
FastEthernet0/0     Up     --     200.1.1.1/24        <not set>                                 00D0.D38B.4101
FastEthernet0/1     Up     --     <not set>           <not set>                                 00D0.D38B.4102
FastEthernet0/1.10  Up     --     172.30.72.1/25      <not set>                                 00D0.D38B.4102
FastEthernet0/1.20  Up     --     172.30.72.129/25    <not set>                                 00D0.D38B.4102
FastEthernet0/1.30  Up     --     172.30.73.1/26      <not set>                                 00D0.D38B.4102
FastEthernet0/1.40  Up     --     172.30.73.65/28     <not set>                                 00D0.D38B.4102
FastEthernet0/1.50  Up     --     172.30.73.81/28     <not set>                                 00D0.D38B.4102
Vlan1               Down   1      <not set>           <not set>                                 00D0.585E.11C7

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Router1
```

Figure 2: Cisco 2811 Router interface status showing FastEthernet0/1 sub-interfaces up/up

4.2 Switch Configuration

4.2.1 Core Switch Configuration

The Cisco 3560 Core Switch was configured with all five municipal VLANs (10, 20, 30, 40, 50). All 6 uplink interfaces were explicitly configured with 802.1Q trunking encapsulation (switchport trunk encapsulation dot1q) and set to trunking mode.

```
Core-Switch>enable
Core-Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1
Fa0/2       on           802.1q         trunking      1
Fa0/3       on           802.1q         trunking      1
Fa0/4       on           802.1q         trunking      1
Fa0/5       on           802.1q         trunking      1
Fa0/6       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005
Fa0/2       1-1005
Fa0/3       1-1005
Fa0/4       1-1005
Fa0/5       1-1005
Fa0/6       1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1,10,20,30,40,50
Fa0/2       1,10,20,30,40,50
Fa0/3       1,10,20,30,40,50
Fa0/4       1,10,20,30,40,50
Fa0/5       1,10,20,30,40,50
Fa0/6       1,10,20,30,40,50

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,10,20,30,40,50
Fa0/2       1,10,20,30,40,50
Fa0/3       1,10,20,30,40,50
Fa0/4       1,10,20,30,40,50
Fa0/5       1,10,20,30,40,50
Fa0/6       1,10,20,30,40,50
```

Figure 3: Core Switch 802.1Q trunk port verification output

```
Core-Switch#
Core-Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/1, Gig0/2
10   Admin                            active
20   Staff                            active
30   VoIP                             active
40   Servers                          active
50   Kiosk                            active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Core-Switch#
```

Figure 4: Core Switch active VLAN database listing VLANs 10, 20, 30, 40, and 50

4.2.2 Access Switch Configuration

Each departmental access switch was provisioned with an 802.1Q trunk uplink to the Core Switch and access ports locked to their respective departmental VLANs.

```
Admin-Switch#show running-config
Building configuration...

Current configuration : 1313 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Admin-Switch
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/2
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/3
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/4
 switchport access vlan 10
 switchport mode access
!
```

Figure 5: Admin Switch running configuration showing FastEthernet access ports assigned to VLAN 10

```
Staff-Switch>enable
Staff-Switch#show running-config
Building configuration...

Current configuration : 1313 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Staff-Switch
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/2
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/3
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/4
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/5
!
```

Figure 6: Staff Switch running configuration showing FastEthernet access ports assigned to VLAN 20

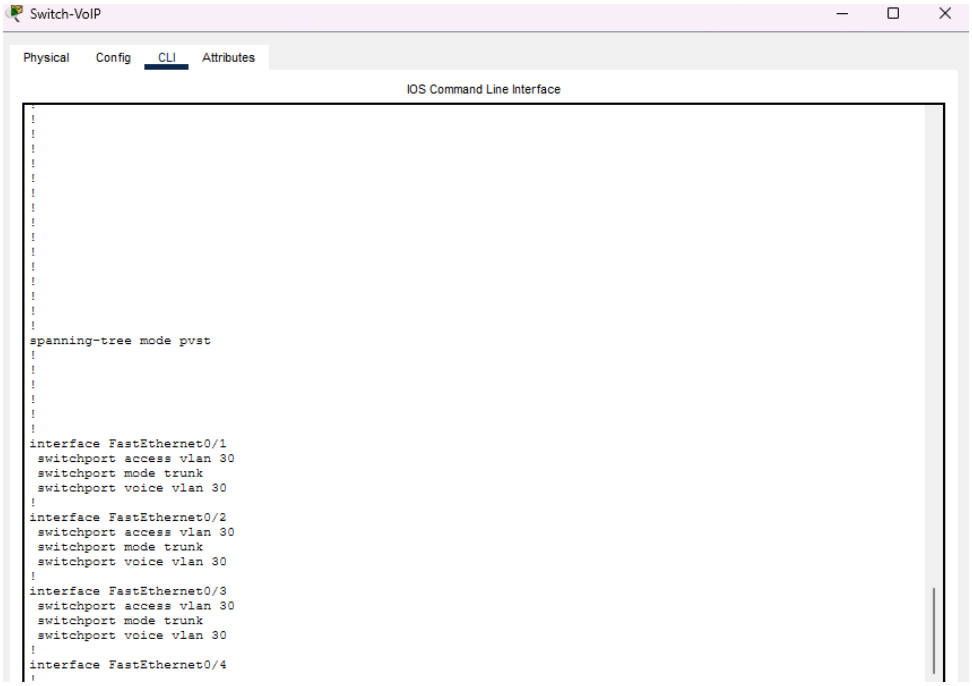


Figure 7: VoIP Switch running configuration showing PoE ports configured with Voice VLAN 30

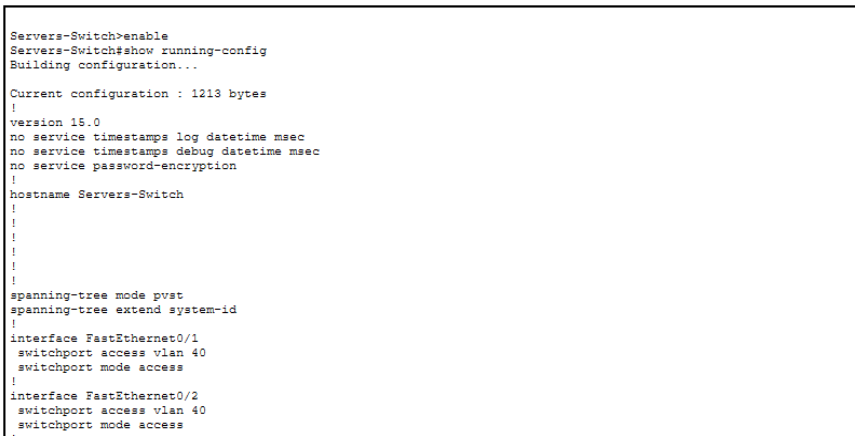


Figure 8: Servers Switch running configuration showing access ports assigned to VLAN 40

```
Kiosk-Switch>enable
Kiosk-Switch#show running-config
Building configuration...

Current configuration : 1160 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Kiosk-Switch
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport access vlan 50
 switchport mode access
!
interface FastEthernet0/2
!
```

Figure 9: Kiosk Switch running configuration showing access ports assigned to VLAN 50

4.3 Server Configuration

The Server-PT device on VLAN 40 (172.30.73.66) was configured with static IP parameters and active DHCP pools for AdminPool, StaffPool, VoIPPool, ServerPool, and KioskPool.

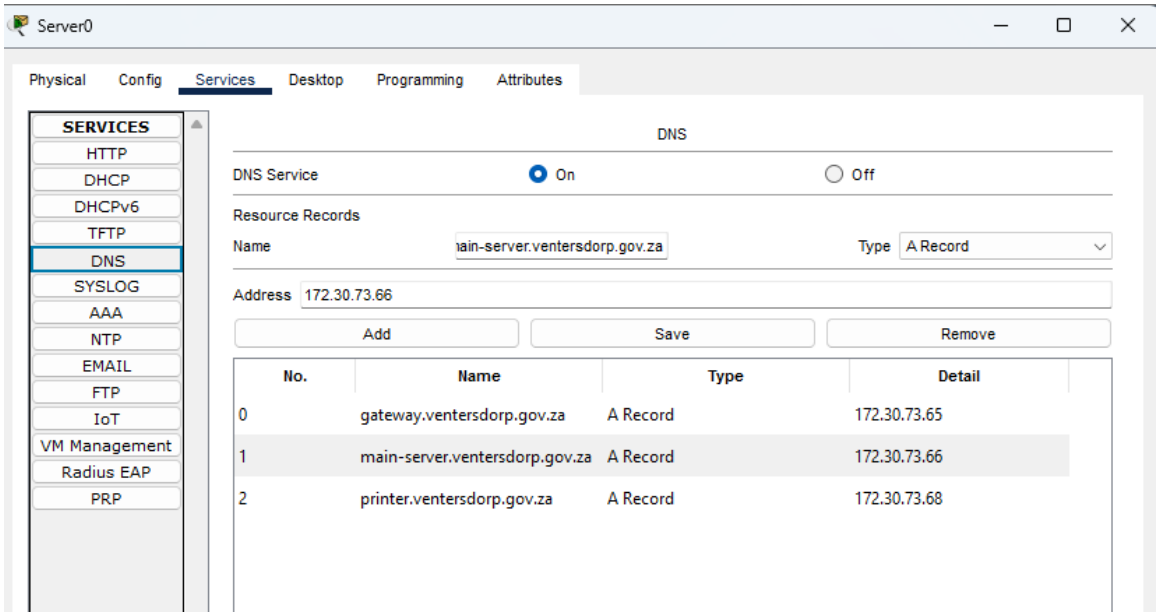


Figure 10: Server DNS service enabled with configured A records list

4.4 VoIP Configuration

Telephony Service (CME) was configured on the Cisco 2811 Router with ip source-address 172.30.73.1 port 2000, ephone-dn extension definitions (1001–1003), and MAC bindings.

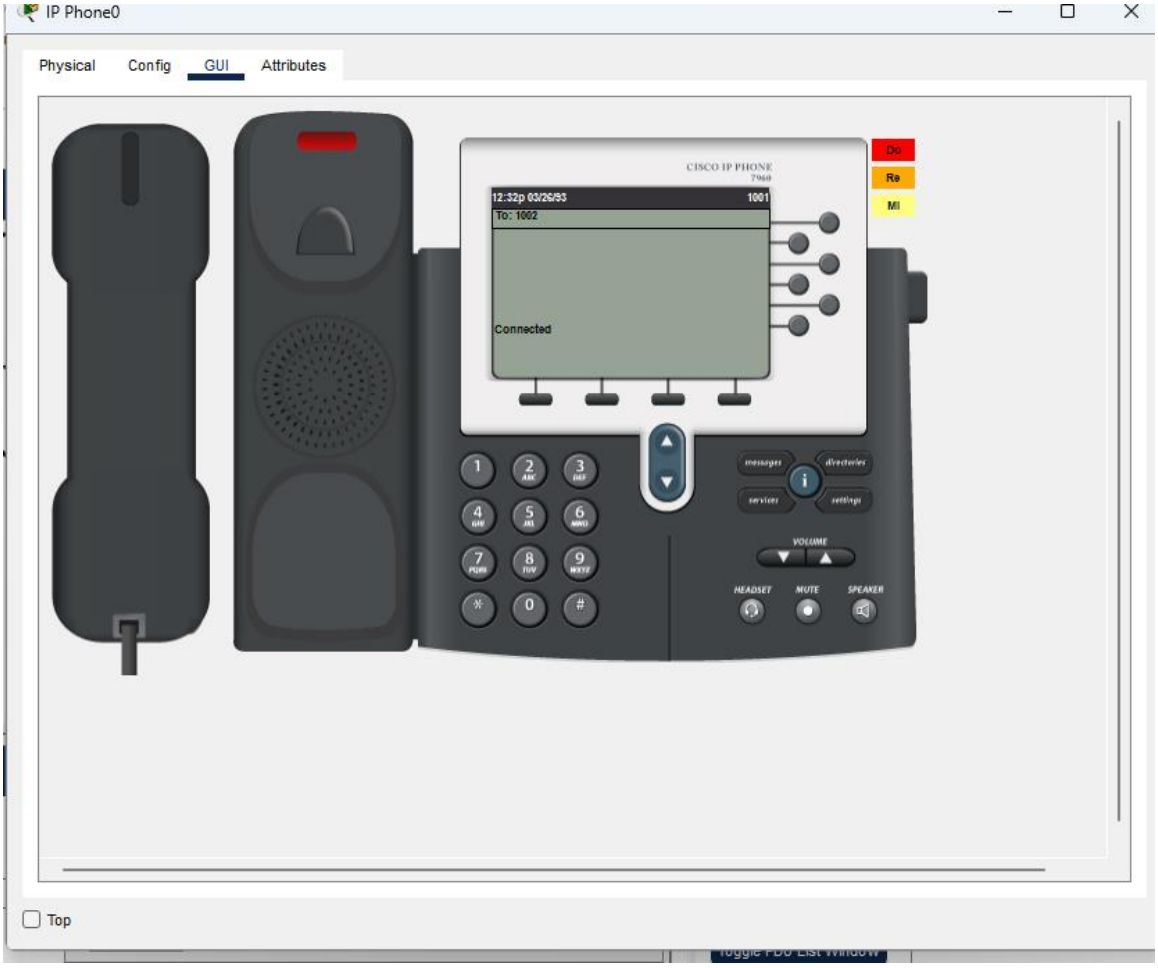


Figure 11: Call Manager Express (CME) configuration and ephone registration status



Figure 12: IP Phone active registration and extension display verification

5. Assigned Challenge: DNS Implementation

5.1 Configuration Details

The networking challenge required establishing an internal Domain Name System (DNS) server to provide municipal hostname resolution across all office subnets.

- **Server IP Address:** 172.30.73.66 (Subnet: 172.30.73.64/28, Gateway: 172.30.73.65)
- **DNS Service Status:** ACTIVE / ON
- **Resolution Scope:** ventersdorp.gov.za zone name resolution

5.2 Configured A Records

The following static A records were configured in the DNS service database:

Hostname / FQDN	Mapped IP Address	Record Type
main-server.ventersdorp.gov.za	172.30.73.66	A Record
printer.ventersdorp.gov.za	172.30.73.68	A Record
gateway.ventersdorp.gov.za	172.30.73.65	A Record

5.3 Verification

Internal DNS resolution was verified using nslookup commands executed from client PCs across VLAN 10 and VLAN 20, confirming accurate IP address resolution.

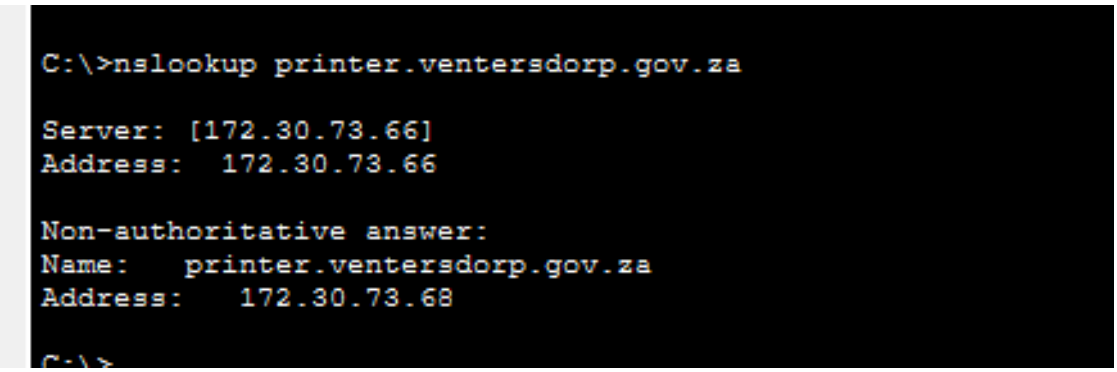


Figure 13: Successful nslookup resolution of printer.ventersdorp.gov.za to 172.30.73.68

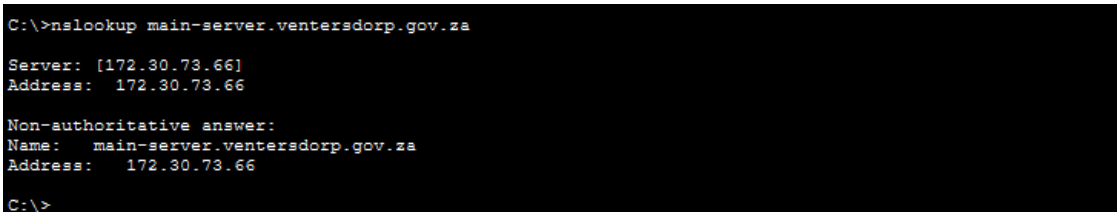


Figure 14: Successful nslookup resolution of main-server.ventersdorp.gov.za to 172.30.73.66

6. Change Request: Public Kiosk (CR13)

6.1 Extended Access Control List (ACL) Implementation

Change Request CR13 mandated deploying a public self-service kiosk at reception providing public internet access while strictly prohibiting access to internal municipal networks. An extended ACL was created and applied inbound on sub-interface FastEthernet0/1.50:

```
Main-Router# show running-config
!
access-list 100 deny ip 172.30.73.80 0.0.0.15 172.30.72.0 0.0.1.255
```

```
access-list 100 permit ip any any
!
interface FastEthernet0/1.50
 encapsulation dot1Q 50
 ip address 172.30.73.81 255.255.255.240
 ip helper-address 172.30.73.66
 ip access-group 100 in
!
```

```
Main-Router#show access-lists
Extended IP access list 100
    10 deny ip 172.30.73.80 0.0.0.15 172.30.72.0 0.0.1.255 (9 match(es))
    20 permit ip any any (11 match(es))
```

Figure 15: Router running configuration snippet showing Access List 100 and sub-interface binding

6.2 Verification

Security compliance was confirmed through sequential testing:

- **DHCP Leasing:** Kiosk PC successfully leases IP 172.30.73.85 via DHCP.
- **WAN Internet Access:** Ping to 200.1.1.1 succeeds with 4/4 replies.
- **Internal Network Isolation:** Ping to internal host 172.30.72.10 times out with 100% loss.

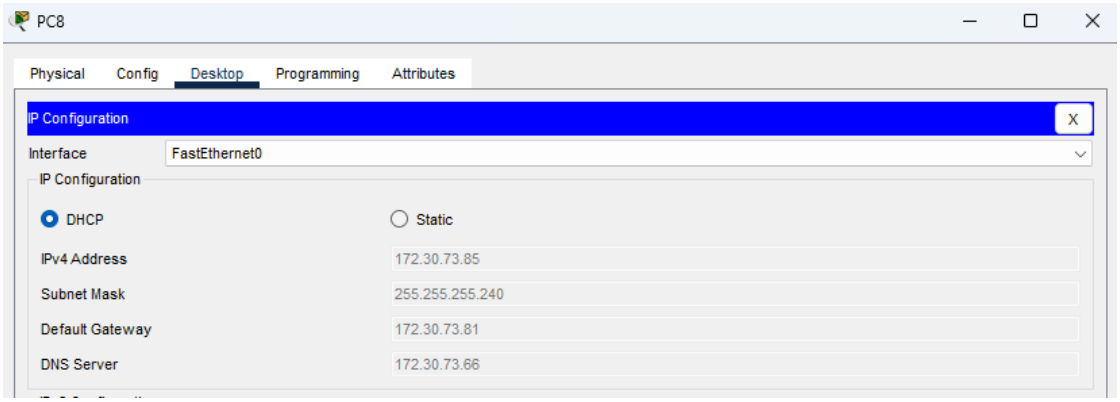


Figure 16: Kiosk PC ipconfig verifying DHCP leasing (172.30.73.85)

```
Pinging 200.1.1.1 with 32 bytes of data:

Reply from 200.1.1.1: bytes=32 time<1ms TTL=255
Reply from 200.1.1.1: bytes=32 time=1ms TTL=255
Reply from 200.1.1.1: bytes=32 time=10ms TTL=255
Reply from 200.1.1.1: bytes=32 time=1ms TTL=255

Ping statistics for 200.1.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 3ms
```

Figure 17: Kiosk PC successful ping to external Internet gateway 200.1.1.1

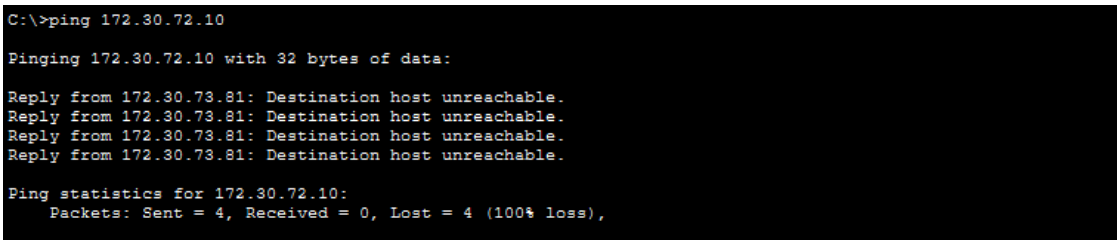


Figure 18: Kiosk PC ping to internal Admin PC (172.30.72.10) blocked by ACL 100

7. Testing & Verification

A comprehensive suite of 9 verification tests was conducted to confirm network operations, routing, name resolution, and security enforcement.

7.1 Test Results Summary

Test #	Test Description	Command	Expected Result	Actual Result	Status
1	Admin PC DHCP Lease	ipconfig	172.30.72.x /25	172.30.72.10	**PASS**
2	Staff PC DHCP Lease	ipconfig	172.30.72.14x /25	172.30.72.141	**PASS**
3	Same-Subnet Connectivity	ping 172.30.72.11	4/4 Replies	4/4 Replies	**PASS**
4	Cross-Subnet Routing	ping 172.30.72.141	4/4 Replies	4/4 Replies (TTL=127)	**PASS**
5	DNS Name Resolution	nslookup printer...	172.30.73.68	172.30.73.68	**PASS**
6	Kiosk PC DHCP Lease	ipconfig	172.30.73.85	172.30.73.85	**PASS**
7	Kiosk Internet Access	ping 200.1.1.1	4/4 Replies	4/4 Replies	**PASS**
8	Kiosk Internal Isolation	ping 172.30.72.10	Destination Unreachable	100% Loss (Blocked)	**PASS**
9	VoIP Phone Registration	show ephone reg	REGISTERED	Ext 1001, 1002 Active	**PASS**

7.2 Additional Test Evidence Screenshots

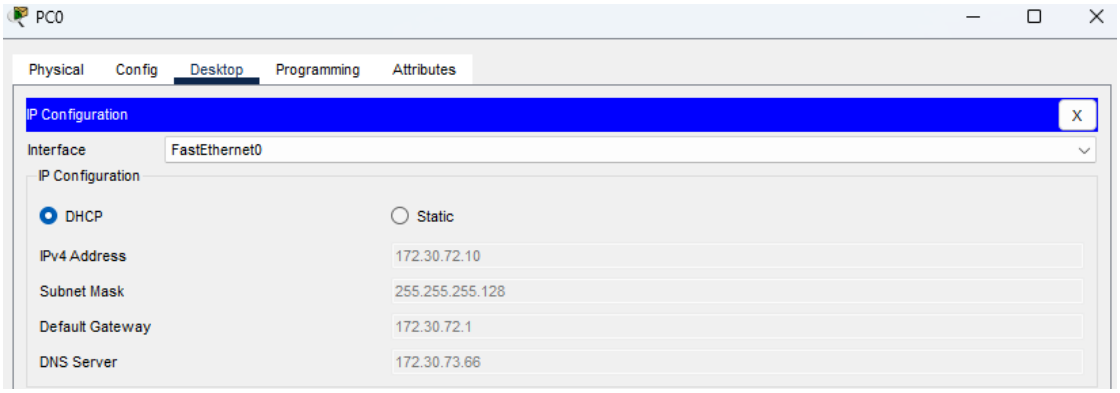


Figure 19: Admin PC ipconfig output verifying DHCP address allocation (172.30.72.10)

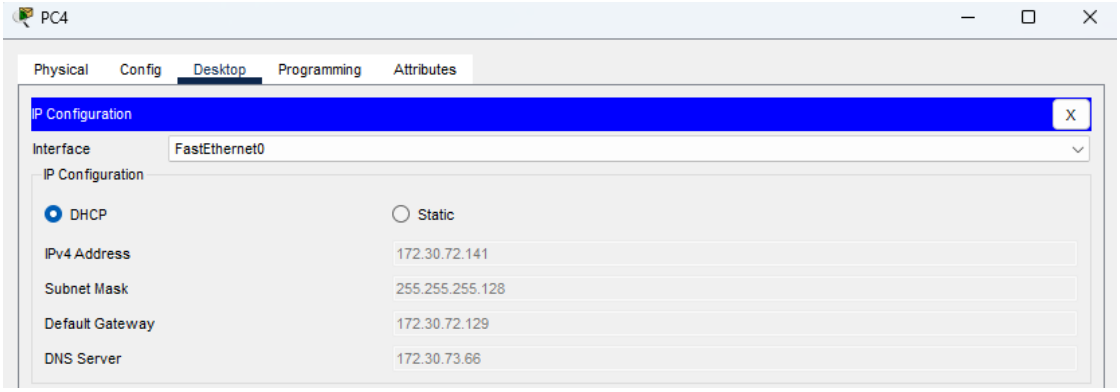


Figure 20: Staff PC ipconfig output verifying DHCP address allocation (172.30.72.141)

```
C:\>ping 172.30.72.11

Pinging 172.30.72.11 with 32 bytes of data:

Reply from 172.30.72.11: bytes=32 time<1ms TTL=128
Reply from 172.30.72.11: bytes=32 time<1ms TTL=128
Reply from 172.30.72.11: bytes=32 time=1ms TTL=128
Reply from 172.30.72.11: bytes=32 time=3ms TTL=128

Ping statistics for 172.30.72.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 1ms

C:\>
```

Figure 21: Same-subnet ping test between Admin PCs (172.30.72.10 to 172.30.72.11)

```
C:\>ping 172.30.72.141

Pinging 172.30.72.141 with 32 bytes of data:

Reply from 172.30.72.141: bytes=32 time<1ms TTL=127
Reply from 172.30.72.141: bytes=32 time=2ms TTL=127
Reply from 172.30.72.141: bytes=32 time=47ms TTL=127
Reply from 172.30.72.141: bytes=32 time=11ms TTL=127

Ping statistics for 172.30.72.141:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 47ms, Average = 15ms
```

Figure 22: Cross-subnet ping test from Admin PC to Staff PC (172.30.72.141) via Router-on-a-Stick

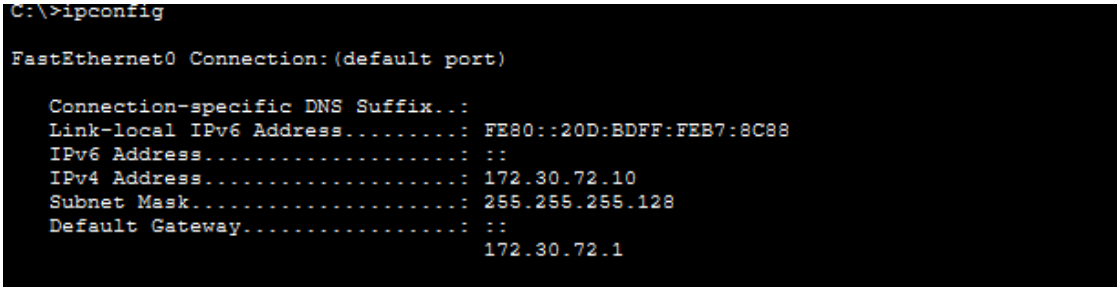


Figure 23: DNS and ICMP end-to-end verification terminal capture

8. Troubleshooting Log

During network implementation, technical challenges were logged, analyzed, and resolved as documented below:

Issue #	Symptom & Error	Root Cause	Resolution Strategy
1	CDP Native VLAN Mismatch errors logged on trunk links.	Core Switch ports configured as access ports while access switches used default Native VLAN 1.	Reconfigured all inter-switch links as 802.1Q trunk ports (switchport mode trunk; switchport trunk native vlan 1).
2	Trunk Encapsulation 'Auto' error on Cisco 3560 switch.	Layer 3 switches require explicit encapsulation definition before enabling trunk mode.	Executed 'switchport trunk encapsulation dot1q' prior to 'switchport mode trunk'.
3	Router failed to ping DHCP/DNS server on VLAN 40.	Servers Switch lacked VLAN 40 definition; ports remained on default VLAN 1.	Created VLAN 40 in switch database and assigned server-facing ports via 'switchport access vlan 40'.
4	PCs received invalid IPs from ServerPool (172.30.73.75).	Default 'serverPool' wildcard mask (0.0.0.0) intercepted DHCP requests prior to departmental pools.	Reconfigured serverPool with specific subnet mask (255.255.255.240) to prevent cross-pool interception.
5	Workstations failed to receive DHCP addresses.	Access switches ran unconfigured factory state without VLAN definitions.	Configured each access switch individually with designated VLANs, trunk uplinks, and access port memberships.
6	IP Phones stuck on 'Configuring IP' / CME registration.	Initial Cisco 2911 model lacked full CME telephony feature support in Packet Tracer.	Replaced gateway with Cisco 2811 running full CME IOS image; updated ephone MAC bindings and Option 150.
7	Kiosk PC initially reached internal servers before ACL enforcement.	Inter-VLAN routing was active without ingress filtering on sub-interface FastEthernet0/1.50.	Applied extended ACL 100 ('ip access-group 100 in' on FastEthernet0/1.50) to block 172.30.72.0 0.0.1.255 traffic.

9. Reflection

Executing this municipal network deployment provided vital insights into enterprise network engineering principles:

- **Traffic Isolation & Quality of Service:** Separating voice traffic on VLAN 30 prevents data packet bursts from causing voice jitter or latency.
- **Router-on-a-Stick Efficiency:** Sub-interface encapsulation on a central router offers a highly efficient, cost-effective routing architecture for satellite branch offices.
- **Granular Security Enforcement:** Applying interface ACLs enforces robust perimeter isolation for public reception terminals at the gateway layer.
- **Infrastructure Services:** Centralized internal name resolution removes the burden of remembering IP addresses from municipal personnel.

Future Recommendations: On future projects, configuration of access switches should occur prior to topology integration, and all debug logs should be systematically recorded in real-time.

10. Conclusion

The network implementation for the JB Marks Local Municipality Satellite Office in Ventersdorp has been fully deployed, tested, and validated. All client requirements have been satisfied:

- ✓ **Topology & Routing:** Hierarchical Star topology with 802.1Q trunking.
- ✓ **Subnet Segmentation:** 5 isolated subnets derived from 172.30.72.0/23.
- ✓ **Challenge Requirement:** Internal DNS zone resolving municipal hosts.
- ✓ **Public Kiosk Security (CR13):** ACL 100 enforced on VLAN 50 granting internet access while blocking internal subnets.
- ✓ **Testing Compliance:** DHCP relay, cross-subnet routing, DNS lookup, and voice registration verified.

11. Appendix A: Device Configurations

Complete Cisco IOS Running Configuration for the Central Gateway (Main-Router):

```
Main-Router>enable
Main-Router#show running-config
Building configuration...

Current configuration : 2035 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Main-Router
!
!
!
!
!
ip dhcp pool VoIPPool
 network 172.30.73.0 255.255.255.192
 default-router 172.30.73.1
 option 150 ip 172.30.73.1
!
!
!
ip cef
no ipv6 cef
!
!
!
!
license udi pid CISCO2811/K9 sn FTX1017D603-
!
!
!
!
!
!
!
!
!
!
!
!
!
!
spanning-tree mode pvst
!
!
!
!
```

```
!
!
interface FastEthernet0/0
 ip address 200.1.1.1 255.255.255.0
 duplex auto
 speed auto
!
interface FastEthernet0/1
 no ip address
 duplex auto
 speed auto
!
interface FastEthernet0/1.10
 encapsulation dot1Q 10
 ip address 172.30.72.1 255.255.255.128
 ip helper-address 172.30.73.66
!
interface FastEthernet0/1.20
 encapsulation dot1Q 20
 ip address 172.30.72.129 255.255.255.128
 ip helper-address 172.30.73.66
!
interface FastEthernet0/1.30
 encapsulation dot1Q 30
 ip address 172.30.73.1 255.255.255.192
 ip helper-address 172.30.73.66
!
interface FastEthernet0/1.40
 encapsulation dot1Q 40
 ip address 172.30.73.65 255.255.255.240
!
interface FastEthernet0/1.50
 encapsulation dot1Q 50
 ip address 172.30.73.81 255.255.255.240
 ip helper-address 172.30.73.66
 ip access-group 100 in
!
interface Vlan1
 no ip address
 shutdown
!
ip classless
!
ip flow-export version 9
!
!
access-list 100 deny ip 172.30.73.80 0.0.0.15 172.30.72.0 0.0.1.255
access-list 100 permit ip any any
!
!
!
!
!
telephony-service
```

```
 max-ephones 10
 max-dn 10
 ip source-address 172.30.73.1 port 2000
 auto assign 1 to 10
 create cnf-files version-stamp Jan 01 2002 00:00:00
!
ephone-dn 1
 number 1001
!
ephone-dn 2
 number 1002
!
ephone-dn 3
 number 1003
!
ephone 1
 device-security-mode none
 mac-address 0060.5CE7.C913
 type 7960
 button 1:1
!
ephone 2
 device-security-mode none
 mac-address 00E0.F746.E6BD
 type 7960
 button 1:2
!
ephone 4
 device-security-mode none
 mac-address 0090.2B42.2253
!
ephone 5
 device-security-mode none
 mac-address 0001.63C5.080D
 type 7960
 button 1:3
!
line con 0
!
line aux 0
!
line vty 0 4
 login
```

12. Deliverables & Project Repositories

The complete source code, Packet Tracer network topology files (.pkt), switch startup configurations, and technical documentation for this project are published and accessible at the repository locations below:

🔗 Official Project Repositories & Submission Links

GitHub Source Repository: https://github.com/siyawnkosi29-debug/cmpg325-municipality-network

Cisco Packet Tracer Topology File (.pkt): *JB_Marks_Satellite_Office_Final_Network.pkt*

Note for Evaluator: The GitHub repository contains the raw .pkt simulation file, complete IOS text configuration files for all 6 switches and the main gateway router, test output logs, and high-resolution topology diagrams.